Industrial Inspection & Maintenance Robot (IIMR)

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Introduction to IIMR

The Industrial Inspection & Maintenance Robot (IIMR) addresses critical challenges in industrial environments, specifically focusing on **safety, downtime, and human error** during inspection processes.

It integrates **robotics, AI, and IoT** to deliver an **autonomous/semi-autonomous and intelligent solution** for comprehensive industrial inspections.

Advanced Sensor Suite

Equipped with **LiDAR, thermal cameras, ultrasonic sensors, and gas detectors** for precise fault detection.

Autonomous Capabilities

Capable of **SLAM-based navigation, obstacle avoidance, and anomaly detection** in complex industrial settings.

Remote Operation

Provides a **remote teleoperation interface** with live video, real-time sensor data, and cloud connectivity for enhanced control.

Industry Alignment

Supports **predictive maintenance** strategies, aligning with key **Industry goals**.

Design and Mobility

AutoCAD Design:

- The robot’s mechanical structure was designed in AutoCAD.
- Provided dimensions and layout for body, wheels, and sensor mounts.
- Helped verify feasibility for industrial inspection tasks.

Inspection Study:

- Analyzed industrial requirements: mobility in tight spaces, real-time sensing, reliable data collection.
- Ensured robot design could operate in **confined and obstacle-rich environments**.

Mobility:

- Selected **differential drive system** (two-wheel drive with caster support).
- Provides stable movement, easy turns, and compatibility with ROS navigation stack.

Sensors:

- LiDAR chosen as the **primary perception sensor**.
- Enables accurate **obstacle detection, mapping, and localization**.

Simulation & ROS Integration

URDF Model Creation

- Built the robot in **URDF (Unified Robot Description Format)**.
- Defined all components: chassis, wheels, sensors, and their connections.
- Ensured compatibility with **ROS** and **Gazebo** for simulation.

Plugins Added

- **Differential Drive Plugin** → Converts velocity commands (`/cmd_vel`) into wheel rotations, and generates odometry data.
- **LiDAR Plugin** → Simulates 2D laser scanning and publishes `/scan` data for obstacle detection.

SLAM Implementation:

- Deployed **gmapping SLAM** within ROS for environment mapping.
- Combined LiDAR scan data with odometry to build **2D occupancy grid maps** in Gazebo.
- Successfully validated robot’s ability to map surroundings and localize itself in a simulated industrial environment.

Custom DWA Local Planner

The Dynamic Window Approach (DWA) is a velocity-based local planner that:

- Generates velocity samples within the robot's dynamic constraints
- Evaluates trajectories based on goal direction, obstacle clearance, and velocity
- Selects optimal velocity commands in real-time

Key Features

- **Python-based implementation** for easy modification and understanding
- **Real-time obstacle avoidance** using laser scan data
- **Smooth trajectory generation** with unicycle model integration
- **Stagnation detection** to prevent the robot from getting stuck
- **RViz visualization** of all evaluated trajectories
- **Goal queue support** for multi-waypoint navigation

Algorithm Parameters

The planner uses a DWAConfig class with the following configurable parameters:

Velocity Limits:

- max_linear_vel: 0.5 m/s - Maximum forward speed
- min_linear_vel: 0.0 m/s - Minimum forward speed
- max_angular_vel: 1.5 rad/s - Maximum turning rate
- min_angular_vel: -1.5 rad/s - Minimum turning rate

Website Work

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What Has Been Done

- Built **Tkinter GUI dashboard .**
- Integrated **live video feed** from onboard camera.
- Implemented and integrated **Up, Down, Left, Right, Stop** teleoperation buttons for manually controlling robot.

Hardware Setup

Hardware Components And Purpose

- **Raspberry Pi 4B –** Raspberry Pi 4B is a compact single-board computer designed for education, prototyping, and embedded applications. It features a quad-core ARM processor, multiple RAM options, and extensive connectivity, making it suitable for robotics, IoT, and general computing tasks.

- **PiCamera2 –** PiCamera2 is the modern camera interface library for Raspberry Pi, built on top of the Linux libcamera framework. It allows Python programs to easily control Raspberry Pi camera modules, capture images/videos, and adjust camera settings with improved performance and flexibility.

- **DC Motors –** DC Motors are electric motors that convert direct current (DC) electrical energy into mechanical rotational motion. They are widely used in robotics and automation because they offer simple speed control, easy interfacing, and reliable performance.

- **BrickPi –** BrickPi is a hardware expansion board that connects LEGO Mindstorms sensors and motors to a Raspberry Pi, enabling advanced robotic control. It allows the Raspberry Pi to act as the programmable brain of LEGO robots, providing more processing power, connectivity, and flexibility than the original LEGO controller.

- **Ngork –** Ngrok is a tunneling service that creates secure, temporary public URLs to your local server, allowing external devices to access it over the internet. It is commonly used for testing webhooks, remote access, and exposing local development environments without deploying them.

Thank You